

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-aa49c38844cbb78bc.jsonl`  
- SHA-256: `39988e520c28c816c39ae14f689713eba569a141c0792f5cad4490b9b930e94b`  
- Source modified: `2026-06-06T17:40:14+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:39:00.138Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Naive audio-only hit detection using raw frame probabilities is poor (F-score ~41-50), but applying the likelihood-ratio confidence measure raises audio-only F-score to roughly 68-77 across four clips ? quantifying how much a simple post-processing trick matters for noisy broadcast audio."

****Source:**** https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information (primary)
****Supporting quote:**** "When using the (absolute) probability of frames, detection performances are quite poor... The use of the likelihood ratio confidence measure significantly reduces the number of false positives in all four video clips."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:39:02.630Z)

I'll verify this claim about audio-only hit detection F-scores. Let me research the source and look for corroborating/contradicting evidence.

3. assistant (2026-06-06T17:39:11.130Z)

I'll verify this claim by examining the primary source and searching for corroborating details. Let me start by fetching the source and searching for the specific numbers.

4. user (2026-06-06T17:39:15.281Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. user (2026-06-06T17:39:29.993Z)

Web search results for query: "Detection of ball hits tennis game audio visual likelihood ratio confidence measure F-score"

Links: [{"title": "Improved Detection of Ball Hit Events in a Tennis Game Using Multimodal", "url": "https://www.isca-archive.org/avsp_2011/huang11_avsp.pdf"}, {"title": "(PDF) Improved detection of ball hit events in a tennis game using multimodal information", "url": "https://www.academia.edu/14129696/Improved_detection_of_ball_hit_events_in_a_tennis_game_using_multimodal_information"}, {"title": "(PDF) Detection of ball hits in a tennis game using audio and visual information", "url": "https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"}, {"title": "Detection of Ball Hits in a"}]

Tennis Game Using Audio and Visual

Information", "url": "http://www.apsipa.org/proceedings_2012/papers/316.pdf"}, {"title": "Detection of ball hits in a tennis game using audio and visual information | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/6411936/"}, {"title": "ISCA Archive - Improved detection of ball hit events in a tennis game using multimodal information", "url": "https://www.isca-archive.org/avsp_2011/huang11_avsp.html"}, {"title": "Detection of Anomalous Events in a Tennis Game Using Multimodal Information", "url": "http://www.apsipa.org/proceedings_2015/pdf/68.pdf"}, {"title": "Wearable Audio and IMU Based Shot Detection in Racquet Sports", "url": "https://arxiv.org/pdf/1805.05456"}]

I've found several research papers on detecting ball hits in tennis games using audio and visual information. Here are the key findings:

Research Overview

Ball hit detection is a key step in understanding a game such as tennis, but single-mode approaches are not very successful: audio detection suffers from interfering noise and acoustic mismatch, video detection is made difficult by the small size of the ball and the complex background of the surrounding environment.

Methodology

The goal is to improve detection performance by focusing on high-level information (rather than low-level features), including the detected audio events, the ball's trajectory, and inter-event timing information. Visual information supplies coarse detection of the ball-hits events, and this information is used as a constraint for audio detection.

Likelihood Ratio and Confidence Measures

The use of the likelihood ratio confidence measure significantly reduces the number of false positives in all four video clips.

Performance Metrics

Combining audio and visual data improves detection metrics across all cases, specifically aiding in accurate identification of volleys and minimizing bounce misrecognitions. After applying this method to a women's doubles tennis game, improvements in the F-score of about 30% (absolute) for audio detection and about 10% for video detection were obtained.

Incorporating inter-event timing significantly refines the detection estimates, as evidenced by improved F-scores in three out of four test video clips.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:39:33.322Z)

The ResearchGate URL is blocked, but I found the actual primary PDF (APSIPA proceedings) and a related paper. Let me fetch the actual PDF to find the exact F-score numbers.

7. user (2026-06-06T17:39:37.949Z)

I cannot extract the requested information about F-scores, precision, recall, and detection performance metrics for audio-only ball-hit detection from the provided web page content.

The page contains general information about APSIPA (Asia-Pacific Signal and Information Processing Association), including details about membership, conferences, publications, and upcoming events. It does not include any technical research data, performance metrics, tables, or findings related to audio-only ball-hit detection systems.

To obtain this information, you would need to access the actual research paper or technical

document that reports these experimental results, which is not represented in this webpage content.

8. user (2026-06-06T17:39:40.735Z)

I cannot extract the requested F-score numbers and specific details about audio-only ball-hit detection from this document. The provided content appears to be a corrupted or encoded PDF file containing primarily binary data and compressed streams rather than readable text about audio detection metrics.

To locate this information, you would need:

1. A properly decoded or text-searchable version of the original document
2. Access to tables specifically labeled for audio-only ball-hit detection results
3. Clear sections discussing confidence measures and probability comparisons

I recommend obtaining the original publication or document in a more accessible format to retrieve these specific quantitative results.

[Binary content (application/pdf, 515.3KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767075390-n4kl6x.pdf]

9. assistant (2026-06-06T17:39:43.532Z)

The ISCA PDF was saved locally as binary. Let me read it directly with the Read tool, and also try the academia.edu version of the multimodal paper.

10. user (2026-06-06T17:39:44.360Z)

PDF file read: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767075390-n4kl6x.pdf (515.3KB)

11. assistant (2026-06-06T17:40:02.148Z)

I now have the primary source data directly from Table 3 of the paper. Let me verify the claim against the exact figures.

The claim states:

1. "Naive audio-only hit detection using raw frame probabilities is poor (F-score ~41-50)"
2. "applying the likelihood-ratio confidence measure raises audio-only F-score to roughly 68-77 across four clips"
3. "across four clips"

Table 3 of the paper ("Detection performance on four video clips using only audio information") gives exact numbers:

****Probability (raw frame probabilities) F-scores:****

- VC(1): 49.56
- VC(2): 49.36
- VC(3): 46.52
- VC(4): 41.23

Range: 41.23 to 49.56 ? matches "~41-50" exactly.

****Likelihood Ratio (LR) F-scores:****

- VC(1): 72.87
- VC(2): 73.02
- VC(3): 67.81
- VC(4): 77.08

Range: 67.81 to 77.08 ? matches "roughly 68-77" (67.81 rounds to ~68, top is 77.08 ? 77).

There are exactly four video clips (VC1-VC4), confirming "across four clips."

The supporting quote is verbatim from the paper's Section 7: "When using the (absolute) probability of frames, detection performances are quite poor... The use of the likelihood ratio confidence measure significantly reduces the number of false positives in all four video clips."

Every numeric range in the claim is confirmed by the primary source's Table 3. The claim is accurately characterized. Let me note one minor nuance: the bottom of the LR range is 67.81, which the claim rounds to "68" ? slightly generous rounding but within "roughly." The "~41-50" is also accurate (actual max is 49.56).

This is a peer-reviewed conference paper (AVSP 2011 / also published APSIPA 2012) from established academic groups (University of East Anglia, University of Surrey), which is appropriate source quality for this descriptive claim. The claim is a factual restatement of the paper's own results table, not an extraordinary claim requiring extra scrutiny.

I could not refute this claim. The numbers, the four-clip count, the "trick matters" characterization, and the supporting quote all check out against the primary source.

12. user (2026-06-06T17:40:10.321Z)

Structured output provided successfully

13. assistant (2026-06-06T17:40:14.441Z)

Verification complete. The claim is NOT refuted ? it is accurately supported by the primary source's Table 3 data, which I read directly from the paper PDF. All numeric ranges (raw probability F-score ~41-50, likelihood-ratio F-score ~68-77, four clips) match the reported figures exactly.